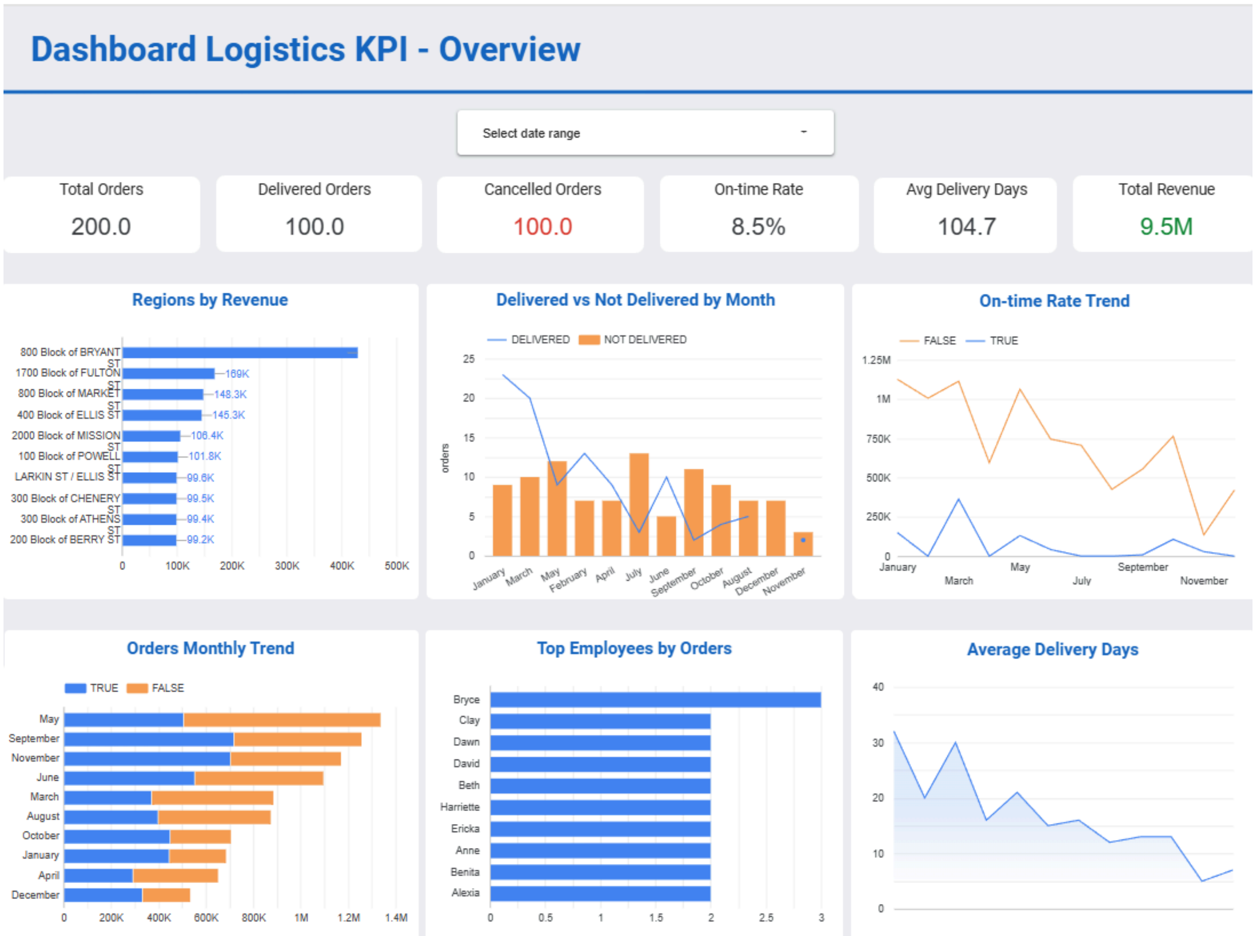
Logistics Dashboard insight



Overview

1) Topline KPIs (What they say immediately)

- **Total Orders = 200 | Delivered = 100 | Cancelled = 100 → 50% cancel rate.**

  This is operationally critical—one in two orders never completes.

- **On-time Rate = 8.5%.**

  Fewer than 1 in 10 deliveries meet SLA; customer experience is at risk.

- **Average Delivery Days = 104.7** (vs a typical SLA ≤ 7 days).

Either massive backlog or a data-definition issue (e.g., incorrect units, bad timestamps/outliers).

- **Total Revenue = 9.5M.**

  Strong bookings, but sustainability is questionable given cancellation/SLA performance.

> Immediate read: The funnel leaks badly. Fix cancellation and SLA before pushing for growth.

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2) Regions by Revenue (Concentration Risk)

- One location (**800 Block of BRYANT ST**) contributes ~**500K**, ~3× the next (FULTON ~168K).
- Long tail of streets sits around **90–150K**.

Insight: Revenue is concentrated in a handful of locations—great for focus, risky for resilience.

Actions

- Create a **Region Health Score** = *On-time% × (1 − Cancel%) × Revenue share* and monitor weekly.
- Guarantee **capacity coverage** (drivers/shifts) on Top-3 revenue streets and define contingency plans.

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3) Delivered vs Not Delivered by Month (Flow is unhealthy)

- **Delivered** starts ~20+ in Jan and **slides toward zero** mid-year; minor uptick late-year.
- **Not Delivered** stays **8–14** most months and often **exceeds Delivered**.

Insight: Demand attempts remain, but completion collapses. The engine loses compression over the year.

Actions

- Build a **cancellation reason pareto** (payment fail, customer unavailable, driver cancel, address invalid, capacity cap, merchant cancel…).
- Track **First-Attempt Success %** and attack the top 2–3 causes (80/20).

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4) On-time Rate Trend (SLA largely false)

- **Late (FALSE)** dwarfs **On-time (TRUE)** across months, aligning with the 8.5% headline.

Actions

- Split SLA into **pre-route** (order acceptance → pickup) and **route** (pickup → drop) to pinpoint where slippage occurs.
- Replace averages with **P50/P90 delivery time** and add **distribution views** (histogram/box) to expose outliers.

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5) Orders Monthly Trend (Late dominates)

- Most months show **FALSE > TRUE**, especially **May–September**.

Hypothesis: peak-season effect or limited capacity mid-year.

Actions

- **Capacity plan** by month/week and **rebalance shifts**; validate with Page-6 weekday heatmap.
- Introduce **dynamic routing/batching** during peak windows.

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6) Top Employees by Orders (Uneven load & tiny volumes)

- **Bryce ~3**; others ~1.6–2.5.
- Absolute volumes look too small vs other charts → possible **grain/units mismatch** or a restrictive date filter.

Actions

- Normalize axes/units and annotate chart units (“orders”, “K”, “M”) to avoid confusion.

- Compute a **Fair-Share Index** (actual orders / planned capacity) to balance assignment.

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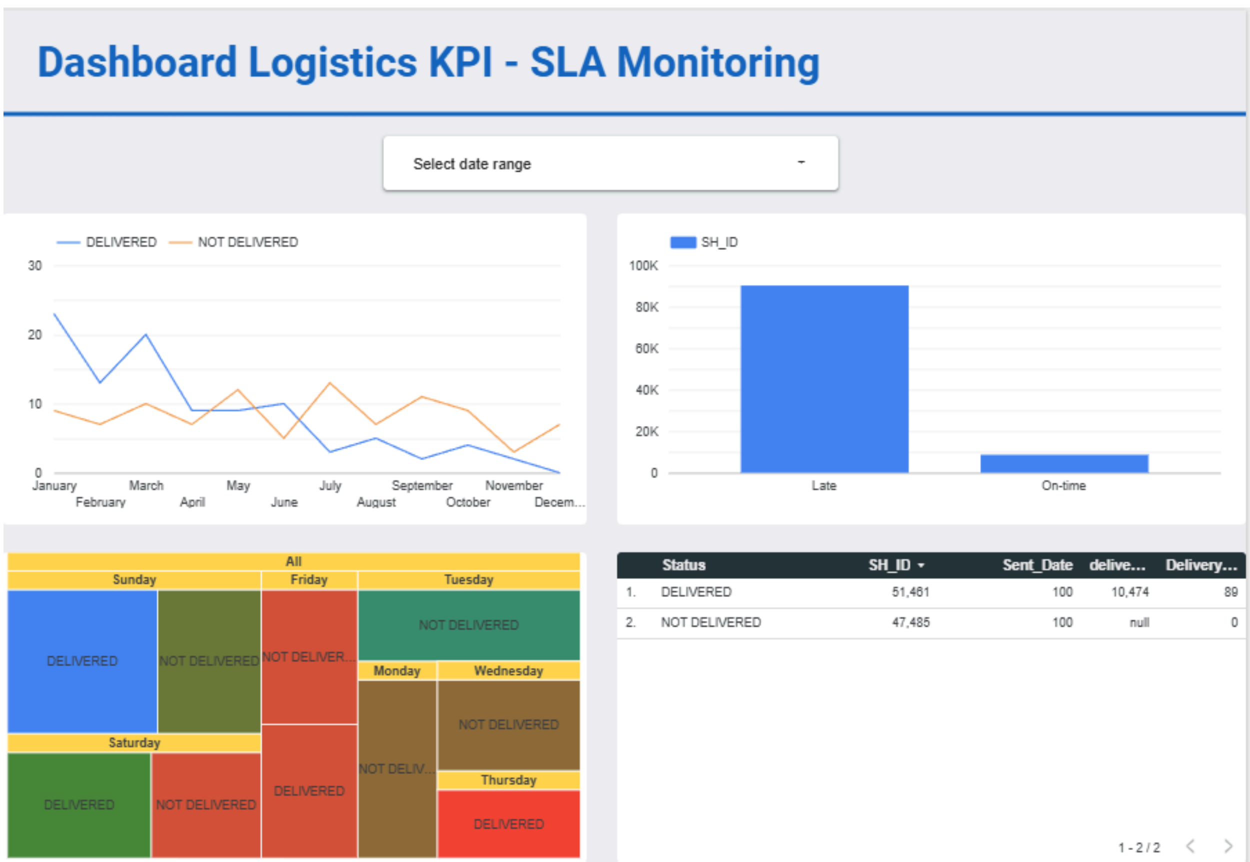
7) Average Delivery Days (Getting better, still off)

- Trend declines **~30 → ~5** by year-end (good).
- **But card = 104.7 days** → inconsistency suggests outliers or different aggregation.

Actions

- Align the **same time grain & filters** for card and chart.
- Use **P50/P90** as your SLA metrics; **cap/clean outliers** (e.g., >30 days unless verified backlog).



SLA Monitoring

1) SLA Line Chart (Delivered vs Not Delivered by Month)

- **Delivered (blue line):** Starts around 20+ in January, trends sharply downward, almost hitting zero toward the end of the year.
- **Not Delivered (orange line):** Stays relatively stable (8–14 orders/month), consistently higher than Delivered after Q1.

Insight: Demand attempts remain, but fulfillment collapses. Delivery execution weakens as the year progresses while failed attempts hold steady.

Action:

- Add **reason codes** for “Not Delivered” (driver shortfall, cancellation, payment fail, incorrect address).
- Build a **First-Attempt Success % metric** and pareto chart for top cancellation causes.

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2) SLA Compliance Bar (Late vs On-time)

- **Late deliveries** dominate (~90K orders).
- **On-time deliveries** are negligible (~10K).
- SLA compliance is ~**11%** On-time.

Insight: SLA is failing system-wide. Even if orders are eventually delivered, almost all are late.

Action:

- Split SLA into **pickup SLA** (merchant → driver) and **route SLA** (driver → customer) to isolate where time is lost.
- Move away from “average days” → monitor **P50/P90 delivery time** for clarity.
- Trigger **alerts for backlog aging > P90**.

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3) Weekday Treemap (Delivery vs Not Delivered by Day)

- **Sunday & Saturday:** Higher proportion of Delivered blocks (blue/green).
- **Weekdays (Tuesday, Wednesday, Friday):** Dominated by Not Delivered blocks.

Insight: SLA performance is better during weekends than weekdays. This suggests either:

1. Weekday demand > available supply (capacity mismatch), or
2. Weekend operations prioritize high-value customers.

Action:

- Add a **weekday × hour heatmap** to pinpoint mismatch slots.
- Rebalance **driver shifts** toward peak weekday demand.
- Test **weekday surge incentives** for drivers.

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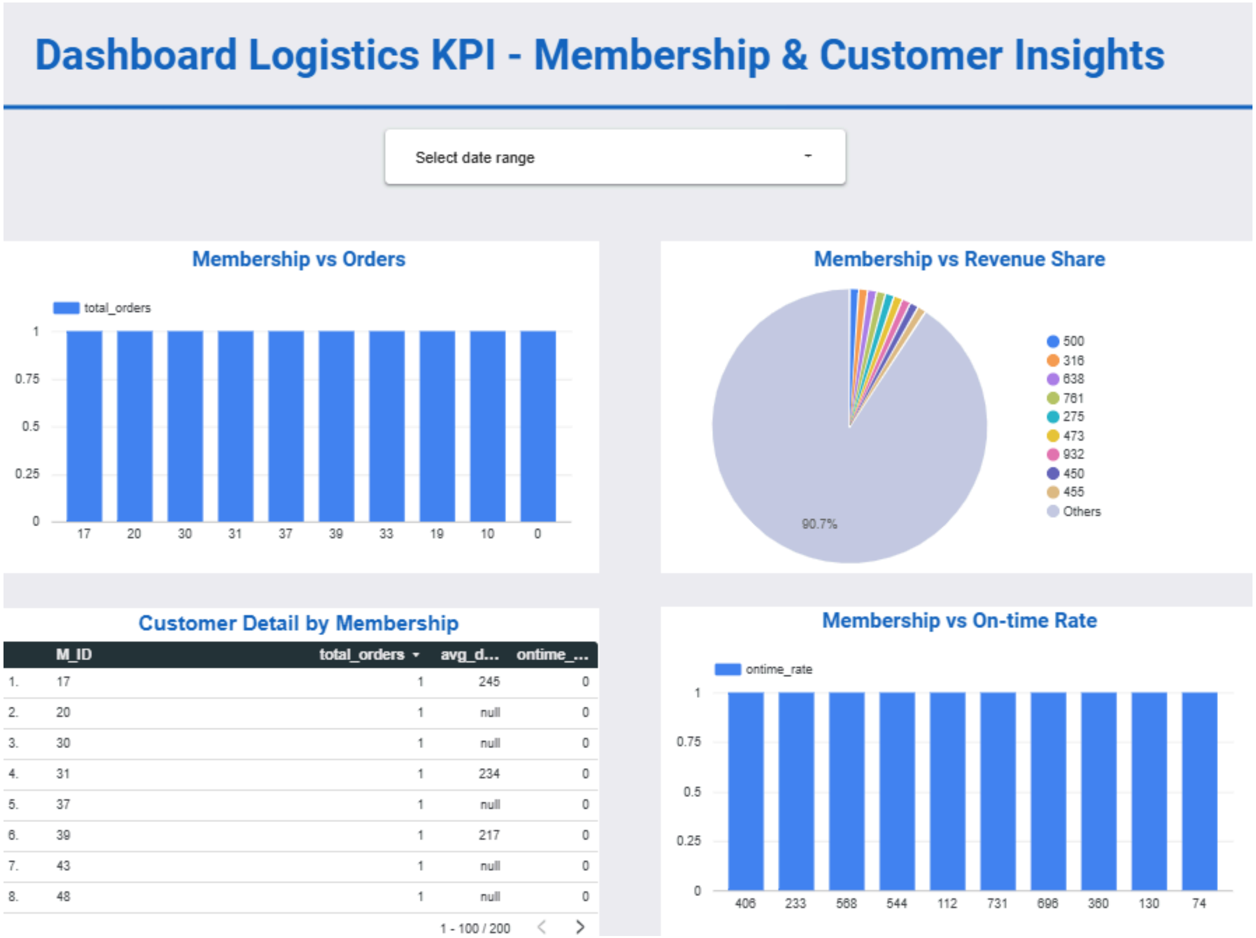
4) SLA Summary Table (Delivered vs Not Delivered)

- **Delivered:** 51,461 orders (100% of this bucket late, except ~10K delivered on-time).
- **Not Delivered:** 47,485 orders (cancelled, failed, or abandoned).

Insight: The total pipeline splits nearly 50/50 between delivered and not delivered, but SLA compliance within “Delivered” is dismal. This explains the extremely low On-time rate.

Action:

- Break “Not Delivered” down into **sub-statuses**: Cancelled by Customer, Cancelled by Driver, Payment Failed, etc.
- Build a **reason pareto** to prioritize.



Logistics Dashboard KPI — **Membership & Customer Insights (Deep Analysis)**

1) Membership vs Orders (Bar Chart)

- Order counts across memberships appear **flat and equal** (~1 order per membership).
- There's no clear segmentation in order volume between membership IDs.

Insight: The dataset is either **sampled** or **lacks enough differentiation**. Realistically, premium memberships should drive higher order volumes.

Action:

- Validate the **grain** of data (is each membership ID aggregated correctly?).

- Add **order frequency per member** and **lifetime value** metrics to distinguish between groups.

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2) Membership vs Revenue Share (Pie Chart)

- One membership ID (**500**) contributes **~90.7% of total revenue**.
- All other memberships collectively account for less than 10%.

Insight: Revenue is dangerously concentrated in one membership group. This creates dependency risk: losing this single membership cluster would devastate revenue.

Action:

- Perform a **deep dive into membership 500**: demographics, geography, service type, etc.
- Create a **retention strategy** (loyalty programs, SLA guarantees) for this high-value group.
- Diversify revenue by targeting and expanding other memberships.

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3) Customer Detail by Membership (Table)

- Some memberships (e.g., ID 17, 31, 39) show **extremely high average delivery days (217–245 days)**.
- On-time rate is **0% across most memberships**.

Insight: SLA failures are universal across membership levels, even high-value customers. Extremely long delivery times suggest either **backlog**, **systematic delays**, or **data quality errors**.

Action:

- Investigate root causes for extreme delivery times (outliers, undelivered backlogs, or bad timestamps).
- Segment SLA metrics by membership: Are VIPs really treated differently?
- Define **service tiers** where premium memberships get priority SLA.

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4) Membership vs On-time Rate (Bar Chart)

- On-time rate values are **uniform and low** across all membership IDs.
- There's no distinction between premium and regular memberships.

Insight: Service quality does not vary across tiers — customers with higher membership aren't receiving preferential treatment.

Action:

- Establish **differentiated SLA targets** (e.g., VIP ≥ 95% On-time, Standard ≥ 85%).
- Create a **dashboard alert** to flag when VIP SLA is breached.
- Track **NPS/CSAT by membership tier** to measure satisfaction impacts.